

We Want Wire-free

What will deliver?

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Sometimes, it just happens that staying behind the technology of the times allows one to skip intermediate technologies and move on to newer ones—at least, that seems to be the case with India. On the cellular communication front, India has not yet moved to 3G from its current 2.5G, and will probably skip it altogether to embrace the emerging 4G. This is seen to be more feasible and 1.5 times more cost-effective, according to Motorola CTO Padmasree Warrior. Though no plans have been announced thus far, we can expect this to happen within the next four years.

The Wireless World Research Forum defines 4G as an Internet technology that combines technologies such as Wi-Fi and WiMAX to enable the lowest-cost wireless network possible. This fourth generation mobile communication protocol aims at delivering wireless broadband access, Multimedia Messaging Service, video chat, mobile TV, high-definition TV content, and DVB (the Digital Video Broadcast standard), in addition to the usual voice services on mobile phones.

Meanwhile, 3G doesn't seem to be as dead as it's been made out to be. Universal Mobile Telecommunications System (UMTS) is a 3G technology that combines 3G and Wi-Fi: it is a high-bandwidth technology that enables CDMA mobile users to experience applications such as videoconferencing, live TV, and Internet connectivity at theoretical speeds of up to 14 Mbps. Practical speeds are found to be up to 3.6 Mbps on compatible handsets, though. Our government is evaluating the process of spectrum allocation for this service, and commercial launch is expected in 2008. This will usher in the era of video phones—read: video chat over your phone. Interesting enough?

4G will use a technology called Ultra Mobile Broadband (UMB), which will use larger bandwidth (greater than that offered by 3G) to deliver a host of Internet and high-bandwidth applications in addition to delivering higher-quality voice services. UMB will help eliminate the disadvantages of CDMA: it will support different and mixed cell sizes (thus allowing more flexibility to operators, who will now be able to create cells according to the number of mobile subscribers) and variable bandwidth sizes that will eliminate the limits of the total bandwidth available to handsets.

In the Internet segment, WiMAX (the 802.16 wireless standard) could well be the next big thing on the wireless communications scene. It offers a considerably longer

range compared to Wi-Fi, and is suitable for wireless broadband. The advantages over traditional wired connections are clear—no wired infrastructure required, low cost of maintenance, and low downtimes—though the initial investment is steeper than that of wired networks. WiMAX will have a wider reach and is well-suited to cover semi-urban and rural India.

WiMAX has had its problems in this country, because the globally-used band of 2.5 GHz–2.7 GHz for WiMAX has been pre-allocated to satellite-based mobile and broadcast transmissions for when national emergencies and natural disasters happen. The government has therefore allocated the remaining bands of 2.3–2.4 GHz to ISPs, and they have to tune their 2.5 GHz certified equipment to this lower bandwidth. This means loss of efficiency. The government has been reluctant to give the higher bands to the ISPs, as these are in use for military purposes. The current government, however, has acted favourably, and slowly but surely, India's highly-regulated wireless spectrum will be opened up.

Intel has spearheaded the “Unwire Pune” programme, which will utilise WiMAX and Wi-Fi to provide high-speed wireless Internet on laptops and PDAs. The chip giant has taken the responsibility of deploying the infrastructure, project management, and the economic management of the entire project, which is still in the pilot phase. The rest of the country might well follow suit—as it was in the case of cable internet—and WiMAX could well become the de facto broadband protocol in India.

Motorola has put forth the concept of Seamless Mobility. This will help extend your connectivity through a host of standards such as Wi-Fi, WiMAX, wireless broadband, and more. One may be attending a video-teleconference at the local WLAN at office, and then move to the traditional cellular environment while driving home—and switch to the fixed network at his home; this will happen seamlessly without any intervention on one's part and without a break in the connection. A prototype has already been demonstrated, but what's required is co-operation between the different agencies such as ISPs and cellular companies, not to mention clearance from the government.

Motorola has already made corporate commitments worldwide and invests a whopping \$3.1 billion annually in this project. No time-frame has been defined for Seamless Mobility, but we're hoping it will happen sooner than later.

So we've told you about what's to come, but you knew it all along—better connectivity, higher speeds, and what-not! What else can possibly happen? ☒

Illustration: Shiksha Parkar